

A Positive Answer to the Hilbert-Space Orlicz-Admissibility Question

Literature status for arXiv:1709.04261

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Status: literature already answered. Question 27 of Jacob–Schwenninger–Zwart has a positive answer, in stronger generality, by Corollary 3.8 and the author-explicit Remark 3.10 of Preußler–Schwenninger, arXiv:2605.12121.

Source question

Jacob, Schwenninger, and Zwart study control systems

$$\dot{x}(t) = Ax(t) + Bu(t), \quad B : U \longrightarrow X_{-1},$$

and ask when L^∞ -admissibility improves to admissibility for an Orlicz space E_Φ , which in turn yields continuity of mild solutions and the equivalence of input-to-state stability (ISS) and integral ISS [1].

Their Section 6, arXiv PDF page 16, Question 27 asks: if A generates an exponentially stable analytic semigroup on a Hilbert space, the H^∞ -calculus of A is not bounded, and $b \in X_{-1}$ is a scalar L^∞ -admissible control operator, must there exist a Young function Φ for which b is E_Φ -admissible?

Exact later answer

Preußler and Schwenninger prove in Corollary 3.8, arXiv PDF page 14:

If X is a reflexive Banach space, then any C -admissible operator is E_F -admissible for some Young function F .

Every L^∞ -admissible operator is C -admissible by restriction to continuous inputs, and every Hilbert space is reflexive. Therefore

$$b \text{ is } L^\infty\text{-admissible} \implies b \text{ is } E_F\text{-admissible for some } F.$$

No analyticity, exponential stability, scalar-input hypothesis, or assumption on the H^∞ -calculus is needed for this implication. Remark 3.10 on the same page explicitly states that source Question 27 is resolved positively by Corollary 3.8 [2]. Thus this is an author-explicit later answer, not an agent-only implication.

Mechanism and wider scope

The key new mechanism is weak compactness. Preußler–Schwenninger use sun duality to represent the relevant adjoint map in L^1 . Weak compactness gives uniform integrability by the Dunford–Pettis theorem; de la Vallée Poussin then supplies a Young function, and Orlicz Hölder yields the extension. Reflexivity makes the continuous-input map weakly compact automatically.

The later paper also gives several broader positive cases for source Question 24: state spaces of nontrivial type, suitable Pełczyński property-(V) hypotheses, and Grothendieck/WCG combinations. Its Corollary 4.7 establishes ISS–integral-ISS equivalence in these classes. Remarks 3.16 and 4.4 address source Questions 28 and 25 in broad settings.

The unrestricted Question 24 for arbitrary Banach state and input spaces is not claimed solved: the later paper explicitly leaves the general ISS/iISS equivalence open outside its geometric cases. The source’s negative L^1 -state-space observation example remains valid.

Verification and provenance

The source’s Question 27 was checked on PDF page 16. The decisive answer was checked on supporting PDF page 14 (Corollary 3.8 and Remark 3.10), and the wider ISS statement on PDF page 20 (Corollary 4.7). The supporting paper cites the 2019 source directly, calls out its numbered questions, and states in its abstract and introduction that it resolves the control-theoretic open problem.

References

- [1] B. Jacob, F. L. Schwenninger, and H. Zwart, *On continuity of solutions for parabolic control systems and input-to-state stability*, J. Differential Equations **266** (2019), 6284–6306; arXiv:1709.04261.
- [2] P. Preußler and F. L. Schwenninger, *Implications of structured continuous maximal regularity*, arXiv:2605.12121v1 (2026).